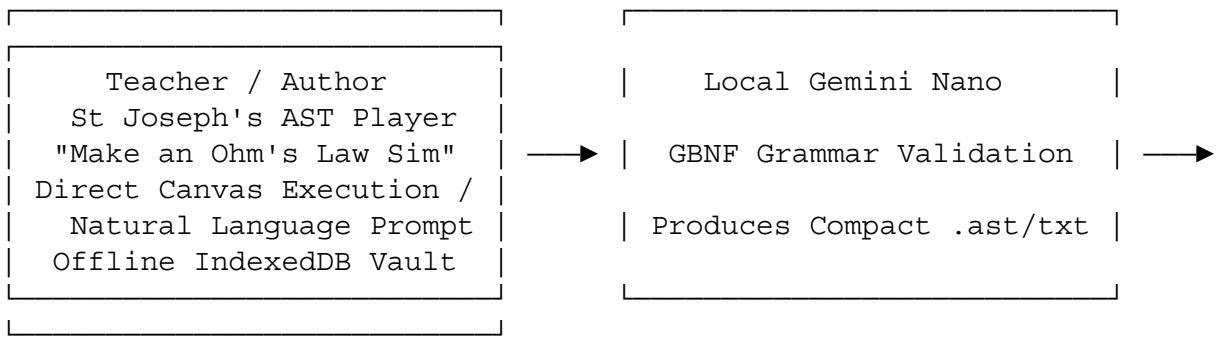
Authoring & Publishing Cartridges via Edge AI



1. How to Prompt Gemini Nano to Generate a Cartridge

Because the engine requires strict declarative syntax, prompts dispatched to Chrome's local window.LanguageModel should feed the core constraints and reference the formal GBNF grammar (/player/grammar/ast-sexpr.gbnf):

The Standard Generation Prompt Template

```
System Prompt:
You are an expert curriculum simulation engineer. You compile
educational apparatuses into inert declarative AST S-Expressions
conforming to the St Joseph's Vector Engine schema.
Do not output HTML, Markdown commentary, or JavaScript. Output ONLY
the raw S-Expression.

Constraints:
1. Root must be: (:scene :id "<id>" :title "<title>" :type
"Sim"|"Slide"|"Film" ...)
2. Target coordinate viewBox: 800 x 480.
3. Every dynamic node must bind via (:bind ... :attr ... :expr ...).
4. Continuous physics MUST use numeric :vars and :dt calculations.
5. All interactive UI controls must be declared inside :inputs with
:min, :max, :step, and :default.

User Prompt:
Create a KS3 Science interactive cartridge demonstrating Archimedes'
Principle:
- A buoyant wooden block submerged in a water cylinder.
- An input slider for fluid density (water vs. oil vs. brine).
```

```
- An input slider for block mass.
- A dynamically updating upward buoyant force arrow and downward
gravity arrow with calculated values.
```

The Chrome Prompt API Call (JavaScript)

```
async function generateCartridge(promptText) {
  // Check Chrome Prompt API availability (Gemini Nano)
  if (!window.ai?.languageModel) {
    throw new Error("Chrome Prompt API (Gemini Nano) not available.");
  }

  const session = await window.ai.languageModel.create({
    systemPrompt: "Output ONLY raw St Joseph AST S-Expressions
matching ast-sexpr.gbnf."
  });

  const response = await session.prompt(promptText);
  return response.trim();
}
```

2. Minimal Valid S-Expression Blueprint

Educators or AI agents can use this baseline template as a scaffolding target:

```
(:scene
  :id "buoyancy-lab"
  :title "Archimedes' Buoyancy Principle"
  :type "Sim"
  :viewBox "0 0 800 480"

  ;; 1. Reactive Input Controls (Auto-HUD discovery)
  :inputs [
    (:slider :id "density" :label "Fluid Density (kg/m³)" :min 800
:max 1200 :step 50 :default 1000)
    (:slider :id "mass" :label "Object Mass (kg)" :min 1 :max 10 :step
1 :default 5)
  ]

  ;; 2. Dynamic State Variables & Physics
  :vars [
    (:let volume 0.008)
    (:let gravity 9.81)
    (:let buoyantForce (* (get "density") (get "volume") (get
"gravity")))
    (:let weightForce (* (get "mass") (get "gravity")))
```

```
    (:let netAccel (/ (- (get "buoyantForce") (get "weightForce"))
(get "mass")))
  ]

  ;; 3. Vector Visual Elements
  :elements [
    ;; Water tank container
    (:rect :x 250 :y 100 :w 300 :h 320 :fill "#e0f2fe" :stroke
"#0284c7" :stroke-width 3)

    ;; Submerged Block with reactive Y coordinate
    (:rect :id "block" :x 350 :y 220 :w 100 :h 100 :fill "#b45309" :rx
6
      :bind [
        (:attr "transform" :expr "(translate 0 (clamp (* (get
'netAccel') -10) -80 80))")
      ])

    ;; Live Force Arrows
    (:line :id "force-up" :x1 400 :y1 220 :x2 400 :y2 140 :stroke
"#16a34a" :stroke-width 5 :marker-end "url(#arrow-green)")
    (:line :id "force-down" :x1 400 :y2 320 :x2 400 :y2 400 :stroke
"#dc2626" :stroke-width 5 :marker-end "url(#arrow-red)")
  ]
)
```

3. How to Upload & Deploy to the Portal

Method A: Client-Side Drag & Drop (Zero Server Footprint)

For immediate classroom use without touching any cloud backend:

1. Open /cartridges (The Decentralized Cartridge Vault) in the browser.
2. Click **"Load Local Cartridge"** or drag and drop a .ast, .astb, or .json file directly into the stage area.
3. The client parses the file locally via src/utils/astBytecode.ts and persists it directly into browser-side **IndexedDB** (services/jotter-db.ts).
4. The custom cartridge immediately appears under **"My Local Lab Cartridges"** and can be run offline on that machine indefinitely.

Method B: Registering in the Production Manifest (School-Wide Distribution)

To make a generated cartridge permanently available across all school boards, kiosks, and LMS iframes:

1. **Save Cartridge File:** Place the compiled file into public/cartridges/ (e.g. public/cartridges/buoyancy-lab.ast).

2. **Register in scenes.manifest.json:** Add the cartridge entry to the central manifest so the HUD and router detect it:
```
{
  "id": "buoyancy-lab",
  "title": "Archimedes' Buoyancy Principle",
  "category": "science",
  "path": "/cartridges/buoyancy-lab.ast",
  "curriculumCode": "OAK-KS3-SCI-014",
  "type": "Sim"
}
```

3. **Optional Binary Pre-compilation (.astb):** Run the offline compression utility to compress the text S-expression into binary bytecode:
`node tools/compile-cartridge.js public/cartridges/buoyancy-lab.ast`
Yields a ~700-byte .astb file ready for instant execution with sub-millisecond parsing.

Why This Unlocks Instant Adoption

- **No Git or Coding Required:** A teacher prompts Gemini Nano locally in natural language, clicks "Test," and watches the simulation boot immediately in the browser.
- **Zero Egress:** The generation prompt, compilation, and storage never touch an external cloud server, keeping student devices strictly compliant with safeguarding and data privacy.